

Module 07: Addressing in Computer Networks

AKTU BCS603 Focus: 4 Addressing tiers (Physical MAC, Logical IP, Port, Specific), address translation protocols, and 3 delivery scopes (Hop-to-Hop vs Host-to-Host vs Process-to-Process).

4 Levels of Network Addressing & Delivery Scope

1. Physical Address (MAC / Link Layer)

- Length: 48 bits (6 Bytes) — Burned into NIC hardware
- Format: Hexadecimal

00:1A:2B:3C:4D:5E OUI (24 bits) Vendor ID NIC Specific ID (24 bits) • Scope: Hop-to-Hop (Node-to-Node) Delivery • Changes at every router hop across the transmission path! • Resolved by ARP (Address Resolution Protocol) 2. Logical Address (IP / Network Layer) • Length: IPv4 = 32 bits (4 Bytes) | IPv6 = 128 bits (16 Bytes) • IPv4 Format: Dotted Decimal 192.168.1.100 Network ID (NetID) Host ID (HostID) • Scope: Source-to-Destination (Host-to-Host) Delivery • REMAINS CONSTANT throughout entire end-to-end journey! • Assigned statically or dynamically via DHCP 3. Port Address (Transport Layer) • Length: 16 bits (0 to 65,535) • 3 Port Categories: 1. Well-known Ports (0 - 1023): HTTP=80, HTTPS=443, SSH=22 2. Registered Ports (1024 - 49151): MySQL=3306, Tomcat=8080 3. Dynamic / Private / Ephemeral (49152 - 65535): Client sockets • Scope: End-to-End (Process-to-Process) Delivery • Socket Address = IP Address : Port Number Identifies specific application process in OS 4. Specific / Application Address • Human-Friendly User-Level Addresses • Web URLs: <https://www.aktu.ac.in/index.html> • Email Address: student@aktu.edu.in • Fully Qualified Domain Names (FQDN) DNS translates Domain Name → 32-bit IP Address • Scope: User / Application Interface Layer Allows humans to navigate networks without memorizing numbers

1. Address Classification & Key Properties

Address Type	Bit Size	Hardware / OS Location	Delivery Horizon
Physical (MAC)	48 bits (6 Bytes)	NIC ROM / Firmware	Node-to-Node (Local subnet only)
Logical (IPv4)	32 bits (4 Bytes)	OS TCP/IP Stack Configuration	Host-to-Host (Global internetwork)
Logical (IPv6)	128 bits (16 Bytes)	OS TCP/IP Stack Configuration	Host-to-Host (Next-gen Internet)
Port Number	16 bits (0 - 65535)	Kernel Socket Table	Process-to-Process (Application binding)
URL / FQDN	Variable ASCII	Application layer / Browser cache	User interface (DNS resolved)

2. The Socket Abstraction

Network communication me process-to-process connection hamesha do **Sockets** ke beech banta hai:

Source Socket: 192.168.1.15 : 54321 (Client Ephemeral Port)

Destination Socket: 142.250.190.46 : 443 (Google Web Server HTTPS Port)

Golden Rule: Source & Destination IP addresses never change as packet moves from router to router. But Source & Destination MAC addresses change at EVERY hop!